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SET-I

Answer..1

Trust management in network security is about figuring out who or what can be trusted inside a network and how much. It is like checking peoples identities before letting them in. Network security trust management involves checking the identity of users, devices and applications before they can access data or services. Think of trust management like a security guard at a gate. Just because someone gets to the gate does not

mean they can come in. They need to show they are supposed to be. For example when you log in to your email or banking app you need to put in a password or a special code. This helps the system know that you are really you. It is like that in company networks too. Only devices that are registered and have good security settings can connect. Some organizations use something called "zero trust" where every user and device has to prove who they are over and over even if they are already inside the network.

Trust management is important because it keeps systems from people who are not supposed to be there and from cyber attacks. Without it bad people could easily get into a network by pretending to be someone they're not. Trust management in network security also helps keep communication, between systems and makes sure that only trusted people can see sensitive data.

The main goals of network security trust management are to keep data safe and make sure communication is secure. These goals are:

1. Confidentiality:
This means that people who are supposed to see information can see it. They use codes to keep data safe. For example when you visit a website your data is coded so others cannot read it. Network security trust management uses confidentiality to protect data.

2. Integrity:
This means that data stays accurate and does not change when it is being sent or stored. If someone tries to change it the system can tell. For example when you download a file the system checks to make sure it is the file and that nobody changed it. Network security trust management uses integrity to protect data.

3. Availability:
This means that systems and data are available when you need them. The system is protected from attacks that could make it stop working. Network security trust management uses availability to keep systems running.

4. Authentication:
This is how the system knows who you are. You use passwords, special keys or codes to prove who you are. Network security trust management uses authentication to verify the identity of users and devices.

5. Authorization:
This is what you are allowed to do once the system knows who you are. For example a normal user can only look at data. An administrator can change or delete it. Network security trust management uses authorization to define what users can do.

Answer..2
In network security, correlation and traceback are really important. They help us figure out what is going on when there is a cyber attack. Correlation is like collecting clues from places. We look at system logs, firewall records and network traffic data to understand what happened. By putting all these clues security teams can see things that might not be obvious at first.

For example if a server has a lot of failed login attempts and the firewall is seeing traffic at the same time correlation helps us connect these things and think that maybe someone is trying to attack us. We use tools

like SIEM systems to do this. Correlation is really helpful because it lets us see patterns and track what the attackers are doing over time. This way we can even catch attacks.

Traceback is different. It is like trying to find out who is really behind an attack. Attackers often try to hide who they are by using tricks like IP spoofing. Traceback helps us follow the traffic back to where it came from. Like in a Denial of Service attack we can use traceback to find out which computers are sending the traffic. Then we can stop them. Prevent more damage.

So correlation and traceback work together to help us catch attackers. They help us see patterns and find out where the attacks are coming from.

Network attacks can be either active or passive. Active attacks are when the attacker tries to change or hurt the system. These are usually easier to catch because they affect how the system works. Examples are Denial of Service attacks, changing data and malware attacks. In a Denial of Service attack the attacker sends much traffic to a server that it cannot handle it.

Passive attacks are when the attacker just listens to the network traffic to get information. These are harder to catch because they do not affect how the system works. Examples are. Traffic analysis. Like if someone's secretly catching data packets to get sensitive information like passwords.

In the end correlation and traceback are crucial, for catching attackers.. Understanding active and passive attacks helps us make better security plans to protect our networks.

Answer..3
Complete mediation is a security principle that means every time a user or a program tries to access a resource the system must check if they are allowed to do. This is very important because it helps keep the people from getting to things they should not have. The system should always check it should not just remember what it checked before. For example when a user opens a file the system should check if they are allowed to read or change the file every time they try to do something with it. This helps keep people from getting to things they should not have especially if the users permissions change or if someone bad tries to use a connection that's already open.

The idea of acceptability in security systems is to make sure the security is easy to use and understand. If the security is too hard to use people might try to avoid it or find ways around it which can be very bad. For instance if the rules for passwords are too hard to remember people might write them down. Use the same password for everything. A good security system should be safe and easy to use at the time. Things like login methods, clear instructions and interfaces that are easy to use can help with this. When people find the security easy to use they are more likely to use it the way, which makes everything safer.

Role-Based Access Control or RBAC for short is a way of controlling access to things based on what role a person has. A role is like a job it has responsibilities. People are given roles. Each role has its own set of permissions. For example in a company someone with the "Admin" role might be able to do anything. Someone with the "Employee" role might only be able to see certain things. Of giving permissions to each person one by one permissions are given to the roles.

RBAC has a lot of things, about it. First it makes things simpler because permissions are given to roles, not people, which makes it less complicated. Second it helps with security because people can only do what they need to do their job, which's a good idea. Third it makes it easier to change permissions when someones role changes. For example if someone gets a job they can just be given a new role instead of changing a lot of permissions. Lastly RBAC helps with keeping track of who can do what which makes it easier to control things.

Overall, mediation, psychological acceptability and RBAC are all very important ideas that help make security systems better and easier to use.

SET-II

Answer..4

Message Authentication Codes are security tools that help verify that a message is real and has not been changed. A Message Authentication Code is a piece of information that is made using a secret key and the message itself. When someone sends a message they also send the Message Authentication Code value. The person who gets the message uses the secret key to make a Message Authentication Code for the message they got and checks it with the one that was sent. If the two values are the same it means the message is real and came from someone we trust.

Message Authentication Codes are very important for communication because they keep data from being changed by someone who should not have access to it. For example when we do banking or use secure messaging systems, Message Authentication Codes make sure that the information sent between users and servers does not get changed while it is being sent. Even if someone bad tries to intercept the message they cannot change it without changing the Message Authentication Code value, which will alert the person who gets the message.

It is also very important for companies to teach their employees about cybersecurity and how to be aware of dangers. This is because people making mistakes is one of the reasons that security gets broken. Employees are often targeted by people who try to trick them with things like fake emails, social engineering and bad software. Teaching programs help employees understand what the common dangers are and how to stay safe. For example employees learn how to tell if an email is suspicious how to make passwords and how not to click on links they do not know. Having teaching sessions also keeps them up to date about new kinds of cyber dangers.

The importance of this kind of teaching is that it reduces the chances of something bad happening and makes the company safer overall. When employees are aware and careful there are successful attacks. It also helps make a culture of security in the company, where everyone's responsible for keeping data safe.. Employees who have been taught can quickly report anything strange which allows the company to respond faster to potential dangers.

There is something called TEMPEST that's about protecting electronic devices from leaking information through electromagnetic signals when computers, monitors or communication devices are working they send out signals that can sometimes be caught and looked at by people to get sensitive information. TEMPEST is the study and control of these signals to prevent access.

For example sensitive systems might be put in rooms that block electromagnetic signals. This makes sure

that secret data cannot be caught from, outside the system.

In conclusion Message Authentication Codes make sure messages are sent safely cybersecurity teaching helps people defend against attacks and TEMPEST protects against data leaks, all of which together make a strong security system.

Answer..5
A buffer overflow attack is a problem for security. It happens when a program puts much information into a special storage area in the memory. This storage area is like a box where the program keeps the information while it is being used.. When the box gets too full the extra information spills over into other parts of the memory. This can hurt the system because it can overwrite things like system information or program instructions.

Attackers use buffer overflow attacks to put code into a system. For example if a program does not check how much information is coming in an attacker can send information that overflows the box and does harmful things. This can let the attacker get into the system without permission crash the system or take control of it. Buffer overflow attacks are a deal for the operating system because they can get around the normal protections and hurt the memory.

To stop these attacks new operating systems use techniques like protecting the memory and making it hard for attackers to find what they are looking for.. If a program is not written well it can still be hurt by these attacks.

There are also programs called malware. Malware is short for software. It is designed to hurt or exploit systems. There are types of malware.

Viruses are programs that attach themselves to files and spread when those files are opened. They can. Delete information and make the system slow.

Worms are like viruses. They can spread by themselves across networks without anyone doing anything. They can use up all the bandwidth. Make the system slow.

Trojan horses look like programs but they have bad things hidden inside. When they are installed they can steal information. Let attackers in.

Ransomware locks up a users information. Demands money to unlock it. This can cause financial and information loss problems.

Spyware watches what a user is doing and collects information like passwords or browsing habits.

Adware shows ads and may watch what a user is doing.

The impact of malware can be very bad. It can lead to information theft, financial loss, system damage and privacy problems. In companies malware attacks can disrupt things. Hurt the companys reputation. Sometimes important information may be lost forever.

In the end buffer overflow attacks hurt the memory and different types of malware cause kinds of problems. Both show how important it is to have security practices, secure coding and regular system updates to protect the operating system and the information. Buffer overflow attacks and malware are problems that can be stopped with good security. Buffer overflow attacks and malware can be prevented by being careful and keeping the system up, to date.

Answer..6
The Address Resolution Protocol is used in computer networks to map an Internet Protocol address to a Media Access Control address. When a computer wants to send data to another computer on the network it needs the Media Access Control address of the destination computer. If it only knows the Internet Protocol address it uses the Address Resolution Protocol to find the corresponding Media Access Control address.

The computer that sends the data broadcasts an Address Resolution Protocol request asking, "Who has this Internet Protocol address?"

The computer with that Internet Protocol address replies with its Media Access Control address.

The sender computer then stores this information in its Address Resolution Protocol cache. Uses it to send data.

The Address Resolution Protocol is simple and fast. It can be vulnerable to attacks like Address Resolution Protocol spoofing, where an attacker sends fake Address Resolution Protocol messages to redirect traffic.

Cryptography plays a role in web security by protecting data from unauthorized access.

It uses techniques like encryption and decryption to ensure that information remains confidential and secure during transmission.

For example when users enter data such as passwords or credit card details on websites cryptography ensures that this data is encrypted before being sent over the internet.

Even if someone intercepts the data they cannot read it without the key.

Cryptography also provides data integrity and authentication through hashing and digital signatures ensuring that the data is not altered and the sender is genuine.

Internet Protocol Security is a set of protocols used to secure communication at the network layer.

It provides confidentiality, integrity and authentication by encrypting and authenticating Internet Protocol packets.

Internet Protocol Security works in two modes: transport mode and tunnel mode.

In transport mode only the data part of the packet is encrypted, while in tunnel mode the entire packet is encrypted and encapsulated within a packet.

Internet Protocol Security is widely used in Virtual Private Networks to create communication channels over the internet.

Secure Sockets Layer, now commonly replaced by Transport Layer Security is used to secure communication at the application layer for web browsing.

It establishes an encrypted link between a web browser and a server.

When a user visits a website, Secure Sockets Layer or Transport Layer Security ensures that the data exchanged is encrypted and protected from eavesdropping.

It also verifies the identity of the server using certificates.

In conclusion the Address Resolution Protocol helps in address resolution within networks cryptography ensures secure data transmission and Internet Protocol Security and Secure Sockets Layer or Transport Layer Security provide protection, for communication making modern networks more secure.